

# PCAP: Programming Essentials in Python

During the Cisco Networking Academy® course, administered by the undersigned instructor, the student has studied the following skills:

- the universal concepts of computer programming (i.e. variables, flow control, data structures, algorithms, conditional execution, loops, functions, etc.)
- developer tools, developer tools and the runtime environment;
- the syntax and semantics of the Python language;
- the fundamentals of object-oriented programming and the way they are adopted in Python;
- the means by which to resolve typical implementation problems;
- the writing of Python programs using standard language infrastructure;
- fundamental programming techniques, best practices, customs and vocabulary, including the most common library functions in Python 3.

This Statement of Achievement acknowledges that during the course PCAP: Programming Essentials in Python, the student has been able to accomplish coding tasks related to the basics of programming, and understands the programming techniques, customs and vocabulary used in the Python language.

By completing the course, the student is now ready to attempt the qualification PCAP – Certified Associate in Python Programming certification, from the OpenEDG Python Institute.

**Ángel Josafat Vázquez Minor**

Student

**Facultad de Ingeniería - UNAM**

Academy Name

**México**

Location

**03/10/2020**

Date

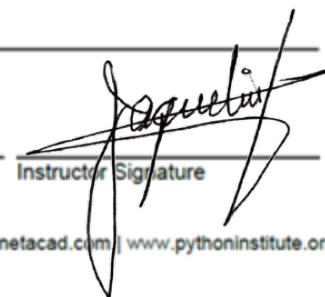
**Dr. Víctor Rangel Licea**

Instructor

  
Instructor Signature

**M.C. Ma. Jaquelina López Barrientos**

Instructor

  
Instructor Signature

